Fixed-target excess: attainment, contact, support and conservation

Independent research note

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Scope of this note

We independently derive the proposed replacement of the extremal ratio by an additive excess over a fixed target. Attainment and closed-support contact hold for every positive target at each fixed index. The conserved identity has the coefficient $(n + H^2)a$, with H the raw sum threshold. This removes the actual score from the contact and remainder coefficients. It does not establish a new uniform local exclusion or automatically accept the old numerical contracts for a different maximizing law. All original source files and certificates are preserved.

1 Definitions and exact scope

Let $\mathcal{P}$ be the laws on $\mathbb{R}$ with $\mathbb{E}X = 0$, $\mathbb{E}X^2 = 1$ and $\rho = \mathbb{E}|X|^3 < \infty$. All sums below use independent copies of one marginal law. For $S_n = X_1 + \cdots + X_n$, set

$$F_{n,P}(H) = \mathbb{P}(S_n \le H), \qquad z = H/\sqrt{n}, \quad R_n(P, H) = \frac{\sqrt{n}}{\rho}\{F_{n,P}(H) - \Phi(z)\}.$$

The global Berry–Esseen constant at this index is the supremum of the absolute standardized discrepancies, divided by $\rho/\sqrt{n}$; call it C_n. Reflection and approximation by closed thresholds show that it is also $\sup_{P,H} R_n(P, H)$. For a fixed number $c > 0$ define

$$a(P, H; c) = \sqrt{n}\{F_{n,P}(H) - \Phi(z)\} - c\rho, \qquad A_n(c) = \sup_{P,H} a(P, H; c).$$

The excess a has the units of ρ times a score difference. It is *not* the existing manuscript's $a_n = C_n - C_{\mathrm{E}}$. At a maximizing pair of this new problem its actual positive score is

$$C_{\mathrm{act}} = c + a/\rho. \tag{1}$$

In general C_{act} need not equal C_n.

For conservation and defect positivity we later specialize to $c = C_{\mathrm{E}} = (3 + \sqrt{10})\phi(0)/6$. Attainment and contact themselves require only $c > 0$. No Fourier bound, numerical acceptance or uniform index limit is part of their proof.

2 Attainment and the equality boundary

Theorem 1. *For each fixed integer $n \ge 1$ and $c > 0$, the maximum $A_n(c)$ is attained by a pair (P, H) with a positive closed-CDF discrepancy. Moreover,*

$$A_n(c) > 0 \iff C_n > c, \qquad A_n(c) = 0 \iff C_n = c, \qquad A_n(c) < 0 \iff C_n < c.$$

Proof. Lyapunov gives $\rho \geq 1$, hence $A_n(c) \leq \sqrt{n} - c$. Symmetric Bernoulli, at $H = n$, gives the stronger lower bound

$$A_n(c) \geq \sqrt{n}\{1 - \Phi(\sqrt{n})\} - c > -c.$$

Write $A = A_n(c)$, and choose $a_j = a(P_j, H_j; c) \to A$. For all sufficiently large j, $a_j \geq A - 1$ and

$$c\rho_j \leq \sqrt{n} - a_j \leq \sqrt{n} - A + 1, \qquad \sqrt{n}\{F_{n,P_j}(H_j) - \Phi(H_j/\sqrt{n})\} = a_j + c\rho_j \geq a_j + c \geq \eta > 0,$$

where η can be any fixed number smaller than $A + c > 0$ once j is large enough. Unit variance implies tightness. A bound $\rho_j \leq M$ gives

$$\int_{|x|>K} x^2 \, dP_j \leq M/K, \qquad \int_{|x|>K} |x| \, dP_j \leq M/K^2.$$

Take subsequences $P_j \Rightarrow P$ and $\rho_j \to b$. Bounded continuous truncation and these tails preserve mean zero and variance one; lower semicontinuity gives $1 \leq \rho(P) \leq b < \infty$.

For $H_j < 0$, Chebyshev gives $F_{n,P_j}(H_j) \leq n/H_j^2$. For $H_j > 0$, the positive discrepancy is at most $1 - \Phi(H_j/\sqrt{n}) \leq n/H_j^2$. Its lower bound $\eta/\sqrt{n}$ thus bounds $|H_j|$. Take $H_j \to H$. Fixed n permits passage from marginal weak convergence to the sum law. For every $t > 0$, eventually $H_j \leq H + t$, so the closed-set inequality and then right continuity give

$$\limsup_j F_{n,P_j}(H_j) \leq F_{n,P}(H + t) \longrightarrow F_{n,P}(H).$$

Consequently

$$a(P, H; c) \geq A + c\{b - \rho(P)\} \geq A.$$

The definition of A forces equality, and $c > 0$ forces $\rho(P) = b$. Third-moment convergence is a *conclusion of maximality*; bounded third moments were not used to assert uniform integrability of third powers. Also $A + c\rho \geq A + c > 0$, proving positive discrepancy.

Reflection changes a negative closed discrepancy into a positive open discrepancy of the reflected law. Closed thresholds increasing to that open threshold approximate it; the penalty is unchanged. This proves the stated supremum identity for C_n without presuming continuity at an atom. If $C_n < c$, every excess is at most $\rho(C_n - c) \leq C_n - c < 0$. If $C_n > c$, a positive closed score above c gives positive excess. If $C_n = c$, choose scores $r_j \uparrow c$ with $r_j \geq c/2$. Their moments satisfy $\rho_j \leq 2\sqrt{n}/c$, and $\rho_j(r_j - c) \to 0$. Thus $A_n(c) \geq 0$, while all excesses are at most zero. These observations prove all three equivalences; no ratio-attainment lemma is needed for this last step. □

3 Atom-safe fixed-target contact

Fix a maximizing pair of Theorem 1, with $n \geq 2$. Set $F = F_{n,P}(H)$, $\varphi = \phi(z)$, $\beta = \mathbb{E}[X|X|]$ and $\gamma = \mathbb{E}X^3$.

Theorem 2. *The following function is nonpositive for every $y \in \mathbb{R}$ and vanishes on the entire closed support of P:*

$$\begin{aligned} I_c(y) = n^{3/2}\{F_{n-1,P}(H - y) - F\} + ny\varphi + \tfrac{1}{2}H\varphi(y^2 - 1) \\ + \tfrac{3}{2}c\rho(y^2 - 1) - c\{|y|^3 - \rho - 3y\beta\}. \end{aligned} \tag{2}$$

That support is compact. Equivalently, for $h(y) = F_{n-1,P}(H - y)$, $h \leq Q_c$ everywhere and $h = Q_c$ on the closed support, where

$$\begin{aligned} nQ_c(y) = nF + \frac{H\varphi + c\rho}{2\sqrt{n}} - \left(\sqrt{n}\varphi + \frac{3c\beta}{\sqrt{n}}\right) y \\ - \frac{H\varphi + 3c\rho}{2\sqrt{n}} y^2 + \frac{c}{\sqrt{n}}|y|^3. \end{aligned} \tag{3}$$

Proof. Contaminate the raw law by $P_\varepsilon = (1-\varepsilon)P + \varepsilon\delta_y$. Its mean and variance are $m_\varepsilon = \varepsilon y$ and $v_\varepsilon = 1 + \varepsilon(y^2-1) - \varepsilon^2 y^2$. Let $r_\varepsilon = \mathbb{E}_{P_\varepsilon}|X - m_\varepsilon|^3$. The standardized law, evaluated at the standardized image of the *fixed raw event*, has excess

$$G_c(\varepsilon) = \sqrt{n}\left\{F_{n,P_\varepsilon}(H) - \Phi\left(\frac{H - nm_\varepsilon}{\sqrt{nv_\varepsilon}}\right)\right\} - c\frac{r_\varepsilon}{v_\varepsilon^{3/2}}.$$

For small $\varepsilon \geq 0$ this is admissible; maximality gives $G_c(\varepsilon) \leq G_c(0) = a$. The raw CDF is a polynomial in ε and has derivative $n(h(y) - F)$. Dominated differentiation of $|x - \varepsilon y|^3$, bounded by $3|y|(|x| + |y|)^2$, gives

$$r_0' = |y|^3 - \rho - 3y\beta, \quad \left(\frac{H - nm_\varepsilon}{\sqrt{nv_\varepsilon}}\right)_0' = -\sqrt{n}\, y - \frac{H}{2\sqrt{n}}(y^2 - 1).$$

Differentiation now gives $G_c'(0+) = I_c(y) \leq 0$. There is no division by ρ in this fixed-target derivative.

Integrating (2) against P gives zero. All its terms are integrable under the third-moment assumption. The map h is upper semicontinuous: at a jump the closed event chooses its larger value. Thus I_c is upper semicontinuous. If it were strictly negative at a closed-support point, a neighborhood of that point would have positive P-mass and a strictly negative upper bound for I_c, contradicting its zero integral. Hence it vanishes at every such point, including atoms and accumulation points. Because $c > 0$, a bound of the form $I_c(y) \leq -c|y|^3 + M_2 y^2 + M_1|y| + M_0$ excludes both sufficiently distant rays from the support. Rearrangement gives (3). □

Neither proof assumes $H \geq 0$. Reflecting a positive closed discrepancy again does not in general preserve that event. The actual orientation must be retained after choosing the maximizing pair.

4 Predecessor support: the extra term has a favorable sign

Theorem 3. *Suppose additionally $C_{n-1} \leq c$. Put $v = n-1$, $s = \sqrt{n/v}$ and $e_n = n(s-1) - 1/2$. For every support point y,*

$$0 \leq \mathcal{H}_n(H, y) + c\{-|y|^3 + \tfrac{3}{2}\rho y^2 + 3\beta y + \rho e_n\} - na, \tag{4}$$

where, with $F_v(x) = \Phi(x/\sqrt{v})$,

$$\mathcal{H}_n(H, y)/n^{3/2} = F_v(H - y) - F_{v+1}(H) + yF_{v+1}'(H) - \tfrac{1}{2}(y^2 - 1)F_{v+1}''(H).$$

Proof. For the same marginal, $F_{n-1,P}(H - y) \leq F_v(H - y) + c\rho/\sqrt{v}$. Substitute this upper bound and $F = \Phi(H/\sqrt{n}) + (c\rho + a)/\sqrt{n}$ in the zero contact equality. The constant moment contribution is $c\rho\{n(s-1) - 1/2\}$, and the new excess contributes exactly $-na$. The Gaussian derivatives give the displayed expression. This proves the claim for each sign of y and every real H. □

If $a \geq 0$, dropping $-na$ gives the old zero-loss support inequality with coefficient c. Therefore the elementary support bounds in current Section 4 apply after their hypotheses are restated as Theorem 2, $C_{n-1} \leq c$, $a \geq 0$, and the same moment caps. At $c = C_{\mathrm{E}}$, the lower numerical multiplier $0.4095 < c$ remains valid only after its cubic coefficient is negative, as in the existing proof. The all-H fourth-order expansion, both ray signs, and frozen moment caps remain necessary. The structural moment bounds are universal bounds for each law's actual score, so (1) with $a \geq 0$ supplies their lower-score premise. This transfer does not re-prove those external structural inequalities or acquire the later conditional moment caps automatically.

5 Independent conservation calculation

This section needs only the explicit contact relations in Theorem 2, not the identity of the actual score with the contact coefficient. Set $\nu = \mathcal{L}(H - S_{n-1})$ and $\tau(t) = \mathbb{E}[X\mathbf{1}_{\{X \ge t\}}]$. Centering makes $\tau \ge 0$, and signwise Fubini gives $\int \tau = 1$ and

$$\int t\tau(t)\,dt = \gamma/2, \quad \int t^2\tau(t)\,dt = M_4/3, \quad \int |t|^3\tau(t)\,dt = M_5/4.$$

Here $M_4 = \mathbb{E}X^4$ and $M_5 = \mathbb{E}|X|^5$. These moments are available because compact support has already been proved. For each bounded complementary gap (u, v) of the closed support, τ is constant, say $\tau_{u,v}$. Define

$$J = \sum_{(u,v)} \tau_{u,v}(v-u)^2,$$

$$J_1 = \sum_{(u,v)} \tau_{u,v} \int_u^v (t-u)t\,dt, \qquad J_2 = \sum_{(u,v)} \tau_{u,v} \int_u^v (t-u)t|t|\,dt,$$

$$L = \sum_{(u,v)} \tau_{u,v} \int_{[u,v)} (t-u)\,d\nu(t), \qquad D = (3+\sqrt{10})\rho - \gamma - 3J.$$

For support in $[-R, R]$, $J \le 2R$, $|J_1| \le RJ/2$, $|J_2| \le R^2J/2$ and $0 \le L \le 2R$. Thus the sums converge absolutely. The gap/defect theorem supplies $D \ge 0$ when that sign is used below; the algebraic identity itself does not require that theorem.

Put $\mathcal{A} = \int \tau(Q_c - h)$. Contact implies

$$h(u) - h(v) = \nu([u, v)) = Q_c(u) - Q_c(v).$$

Integrating $h(t) = h(v) + \nu([t, v))$ across the gap gives

$$\int_u^v (Q_c - h) = \int_u^v (t-u)(-Q_c'(t))\,dt - \int_{[u,v)} (t-u)\,d\nu(t).$$

The half-open upper endpoint is essential. The slack is zero on the support and $\tau = 0$ outside its convex hull. Substituting Q_c' and summing gives the first evaluation

$$\mathcal{A} = \frac{\varphi J}{2\sqrt{n}} + \frac{G_c}{n^{3/2}} - L, \quad G_c = \tfrac{3}{2}c\beta J + (H\varphi + 3c\rho)J_1 - 3cJ_2. \tag{5}$$

Let X^* have density τ, independently of S_{n-1}. Signwise Fubini gives $\mathbb{E}g'(X^*) = \mathbb{E}[Xg(X)]$. Conditional use of that identity and exchangeability show that $(S_{n-1} + X^*)/\sqrt{n}$ is the zero-bias law of $Z = S_n/\sqrt{n}$. Its CDF at z equals

$$T = \mathbb{E}[(Z^2 - zZ)\mathbf{1}_{\{Z \le z\}}], \qquad \int \tau(t)h(t)\,dt = T.$$

The factor $Z^2 - zZ$ makes endpoint atoms harmless. Integrating the four coefficients of (3) gives

$$\mathcal{A} = F - \frac{\gamma\varphi}{2\sqrt{n}} + \frac{H\varphi(3 - M_4)/6 + cK_q}{n^{3/2}} - T, \quad K_q = \frac{M_5}{4} - \frac{\rho M_4}{2} - \frac{3\beta\gamma}{2} + \frac{\rho}{2}. \tag{6}$$

Multiplying the support contact equality by X and using exchangeability also gives, for $M = \mathbb{E}[Z\mathbf{1}_{\{Z \le z\}}]$,

$$M + \varphi = -\frac{\gamma z\varphi}{2\sqrt{n}} + \frac{cK_1}{n}, \quad K_1 = \mathbb{E}[X|X|^3] - 3\beta - \tfrac{3}{2}\rho\gamma. \tag{7}$$

Define $q_z(x) = (x - z)|x - z|$, and let Z_0 be standard normal. Set

$$\varepsilon = \mathbb{E}q_z(Z) - \mathbb{E}q_z(Z_0) - \frac{2\gamma\varphi}{3\sqrt{n}}.$$

The pointwise hinge identity gives

$$T - \Phi(z) = -\varepsilon/2 - \frac{\gamma\varphi}{3\sqrt{n}} + z(M + \varphi) - z^2(F - \Phi(z)).$$

Substitute (7) and $\sqrt{n}(F - \Phi) = c\rho + a$ in (6). Direct collection gives

$$\begin{aligned}\sqrt{n}\mathcal{A} &= c\rho + a - \gamma\varphi/6 + P_0/n + \sqrt{n}\varepsilon/2,\\ P_0 &= H^2(c\rho + a + \gamma\varphi/2) - H\{cK_1 + \varphi(M_4 - 3)/6\} + cK_q.\end{aligned}$$

Equating with (5) proves the following identity.

Theorem 4. *Let $U = J_1 + (M_4 - 3)/6$ and define*

$$\begin{aligned}\mathcal{R}_c(H) = c\{3\beta J/2 + 3\rho J_1 - 3J_2 - K_q\} + H(cK_1 + \varphi U)\\ - H^2(c\rho + \gamma\varphi/2) + E_n, \qquad E_n = -n^{3/2}\varepsilon/2.\end{aligned}$$

Then, for general $c > 0$,

$$(n + H^2)a + nc\rho - \frac{n\varphi}{6}(\gamma + 3J) + n^{3/2}L = \mathcal{R}_c(H). \tag{8}$$

In particular, at $c = C_{\mathrm{E}}$ and $w = H^2/(2n)$,

$$\boxed{\begin{aligned}(n + H^2)a + nc\rho(1 - e^{-w})\\ + \frac{n\phi(0)e^{-w}}{6}D + n^{3/2}L = \mathcal{R}_c(H).\end{aligned}} \tag{9}$$

If $a \geq 0$ and the physical defect inequality $D \geq 0$ holds, all four terms on the left are nonnegative.

The na term comes from the ordinary discrepancy, and H^2a comes from the quadratic hinge. Replacing their sum by na in the exact identity is false; dropping H^2a gives a valid weaker inequality when $a \geq 0$. The formula uses the raw H; in standardized units the coefficient is $n(1 + z^2)$.

5.1 What the Fourier representation still requires

The unchanged compensated-inversion lemma is the exact retained input at manuscript Section 5b, lines 143–200. It applies to a finite signed measure of zero mass and first moment with finite absolute second moment, for every real raw threshold. Applied to the sum law minus the Gaussian and cubic signed comparator, it gives

$$E_n = \frac{2\sqrt{n}}{\pi}\int_0^\infty \frac{\Im(e^{-itH}V_n(t))}{t^3}\,dt, \qquad V_n(t) = f(t)^n - e^{-nt^2/2}(1 - in\gamma t^3/6).$$

The signed comparator has mass one and first moment zero. Compact support already gives all required moments. No occurrence of the contact coefficient or the extremal score is present in this inversion step, so the existing proof applies without modification. This identity does not bound the integral. Complete low- and high-frequency envelopes, signed phase uniformity, domain coverage, moment gates and the infinite tail remain obligations. No nonlattice decay is available merely from the fixed-target reformulation.

6 Gaussian weight and phase price

Corollary 5. *Suppose $c = C_{\mathrm{E}}$, $a \geq 0$, $D \geq 0$, $n \geq N \geq 2$, and an independently established raw upper bound gives $\mathcal{R}_c \leq r_0 < 2N/5$. Then*

$$(n + H^2)a + \frac{n\phi(0)}{6}D + n^{3/2}L \leq \mathcal{R}_c.$$

Any later upper bound $\mathcal{R}_c \leq r$ consequently gives $e^{-w} \geq 1 - 5r/(2n)$ and $nD \leq 15000r/997$, using the same directed constants as the manuscript. If also $nD \leq Nd_D$ and $0.4095\rho - d_D/15 > 0$, set

$$q_{\mathrm{extra}} = \frac{1 - 5r/(2N)}{2}(0.4095\rho - d_D/15) > 0 \quad \textit{when } r < 2N/5.$$

A valid signed envelope $\mathcal{R}_c \leq A(c,\rho) + F_{\operatorname{sgn} H}(c,\rho)|H| - Q(c,\rho)H^2$ implies, for $\kappa = 997/15000$,

$$(n + H^2)a + \kappa nD + n^{3/2}L \leq A(c,\rho) + F_{\operatorname{sgn} H}(c,\rho)|H| - (Q + q_{\mathrm{extra}})H^2.$$

Proof. Equation (8) gives

$$\frac{\varphi}{6}(\gamma + 3J) = (1 + H^2/n)a + c\rho + \sqrt{n}L - \mathcal{R}_c/n > c\rho - 2/5 > 0.$$

Therefore $c\rho - \phi(0)D/6 > 0$. Regrouping (9) leaves the additional nonnegative term $n(1 - e^{-w})(c\rho - \phi(0)D/6)$, which may be dropped. The density and defect bounds follow by retaining one nonnegative term at a time and using $c > 2/5$, $\phi(0) > 997/2500$. Finally $1 - e^{-w} \geq we^{-w}$ and $D \leq d_D$ price that same term by $q_{\mathrm{extra}}H^2$. It is subtracted once, not counted twice. □

7 Transport of a zero-barrier scalar schedule

There is a useful distinction between the actual score C_{act}, the contact multiplier, and the deleted-sum constants. They coincide or are compared in the old derivation, but their roles can be separated. In this note the contact multiplier is always c. The predecessors remain global bounds for actual deleted convolutions. The actual score is given by (1); it is not set equal to c.

Theorem 6 (Conditional scalar-schedule transport)**.** *Set $c = C_{\mathrm{E}}$ and $N \geq 3$. Consider a finite, ordered, zero-barrier acquisition and exclusion schedule at every actual $n \geq N$. Suppose its proof has the following properties.*

1. *Its primitive moment, geometry, Fourier, count and concentration inequalities hold for every actual law in their stated domains. Its raw remainder estimates are universal scalar inequalities for the displayed $\mathcal{R}_t$, with a positive formal multiplier t in an explicitly covered interval containing c.*

2. *Its support steps use the zero-loss contact bound; its resource steps use the conserved identity, licensed weight removal, phase price, and nonnegativity of the excess. All moment/support domains are acquired in their stated order for the same law. Every denominator, sign, strict gap and uniform-index gate is checked.*

3. *A charged corner bound, if used, subtracts only a nonnegative charge $\chi(t)\rho$ with $\chi(c) = 0$, and its proved scalar envelope contains the point $t = c$. No positive lower excess barrier is used unless it has separately been established for this new witness.*

4. *The confinement proof uses the displayed derivative of the contact majorant with multiplier t and separate valid deleted-sum bounds. Its strict budget remains a strict upper envelope for $\mathcal{R}_c$ after the phase price.*

5. *Every positive-noise leaf strictly bounds the count right side with repair multiplier t on its domain, and its bound at $t = c$ is negative. The complete cover and the zero-noise strict binomial case are independently valid. On every positive-noise count leaf a separately established binomial bound gives $C_B \le c$.*

Then the same scalar bounds and unchanged numerical targets, if proved on those domains, exclude a maximizer of $A_n(c)$ with $a \ge 0$ under $C_{n-1}, C_{n-2} \le c$. An upper bound on C_{act} or on na/ρ is not an entry requirement for this transported schedule.

Proof. Initialize with Theorems 1–3 and the universal moment bounds. For a raw scalar inequality specialize its formal multiplier to $t = c$, without changing the actual marginal or any of its other parameters. Theorem 4 and Corollary 5 supply the required nonnegative resources; discarding $(n + H^2)a$ only weakens the resulting inequality. At a charged step the subtracted formal charge is zero. Thus the same endpoint enclosure dominates the specialized remainder. Support improves by the extra $-na$, which can be discarded. The explicit confinement derivative is the derivative of (3), with coefficient c, whereas the concentration estimates use C_{n-1}, C_{n-2} independently. A strict raw envelope remains strict after discarding any of these nonnegative terms. Induction over the finite ordered schedule establishes every successive enclosure for this law. Nothing is inferred from an unproved fixed point.

For the final count step, let Q be the difference from the binomial CDF and $f(u)$ its normal phase correction, as in the current count identity. Direct subtraction yields

$$\begin{aligned} a + \rho_B(c - C_B) = \sqrt{n}\pi_k\{Q/\pi_k - f(u)\} \\ + \sqrt{n}\left\{\Phi\left(\frac{\delta}{\sqrt{nv}}\right) - \Phi\left(\frac{\delta}{\sqrt{n(v+s)}}\right)\right\} + c(\rho_B - \rho). \end{aligned} \tag{10}$$

The right side is exactly the old comparison right side with formal multiplier $t = c$. Its strict negative upper bound contradicts $a \ge 0$ and $C_B \le c$. The nearest-count mass sign and whole-count curvature proofs therefore transport unchanged once their right-side inequalities are proved. At zero noise, maximizing over H for a fixed law still puts H at a binomial atom, because the penalty is independent of H; the separately assumed strict binomial comparison gives $a < 0$. □

Specific scalar bounds that meet the multiplier test. The continuous-cell atlas bounds the constant, linear and quadratic coefficients of $\mathcal{R}_t$ before using conservation. Its separate convexity in t and ρ is purely algebraic. Thus its four-corner bound remains valid at $t = c$ whenever the scalar interval contains c; it does not require C_{act} to lie in that interval. The two-interval bound in current Section 6 likewise still bounds $G(c)$: with $C_0 < c < \chi < C_1$,

$$G(c) \le \max\{G(C_0), G(\chi)\} \le \max\{G(C_0), G(\chi), G(C_1) - N(C_1 - \chi)\}.$$

This uses only the old low interval and its convexity, not an upper bound for C_{act}. The coherent six-corner bound also contains $c < 0.40974$, where its charge $n(t - 0.40974)_+$ is zero. The fixed-target calculation can eventually evaluate $G(c)$ directly, but the larger old bound remains sound if its primitive input bounds have been proved.

The signed cubic repair uses only $0 < c_* \le t \le c^*$ as a scalar multiplier condition. Setting $t = c \in (0.4, 0.4226)$ keeps both signs of the cubic coefficient valid; the negative case still uses its upper multiplier. No actual-score cap is needed for that version of the repair. The confinement multiplier interval $(0.4, 0.4253]$ also contains c; its two deleted-sum caps instead follow from the first-failure hypothesis.

Limits of this transfer. A scalar enclosure originally justified only at a ratio optimizer is not universal merely because it is stored as numbers. Its derivation must have the properties in Theorem 6. In particular, the complete low/high-frequency and floor-42 domains, the initial raw ceiling licensing weight removal, each strict budget, all acquired moment intervals, and the complete count cover are still inputs. Positive-barrier profiles with $K > 0$ or coefficient intervals excluding c are not covered. No assertion here accepts the old numerical predicates or all retained analytic candidates. This is a proved conditional transport, not an unconditional LOCAL42 theorem.

8 A replacement implication, not an established exclusion

Corollary 7. *Fix $N \geq 3$ and $c > 0$. Assume $C_j \leq c$ for $j < N$. Assume independently that at every $n \geq N$ with $C_{n-1} \leq c$ (and $C_{n-2} \leq c$ if an argument uses that deletion), no maximizer of $A_n(c)$ has $a \geq 0$. Then $C_n < c$ for every $n \geq N$.*

Proof. Choose the first failure of the strict conclusion. All required predecessors are at most c. Theorem 1 provides the forbidden excess maximizer with $a \geq 0$. □

This new exclusion has no upper-score or upper-excess entry premise, so the one-step recurrence is unnecessary *for this conditional implication*. The old LOCAL42 assumption explicitly has both such premises and concerns a ratio maximizer. It cannot be substituted for the new assumption. Exclusion only of $a > 0$ yields a weak global bound, not the strict tail.

For a binomial comparator of score C_B and third moment ρ_B, direct subtraction gives $a = \rho_B(C_B - c) + \mathcal{E}_c$. The bound $\mathcal{E}_c \leq \rho_B(c - C_B)$ yields only $a \leq 0$. Strict exclusion requires a strict comparison or a separately proved strict margin. This equality boundary must not disappear during the reformulation.

9 Adversarial checks and remaining scope

The separate exact checker uses sparse Laurent polynomials over rational numbers to derive the zero-bias moments of Q_c, the truncated first moment, both slack expansions and the hinge elimination. A mutation using na leaves exactly $H^2 a$. Finite-law checks use symmetric and skew Bernoulli, its reflection, a three-point law with a zero atom, a four-point law, and three symmetric tiny-far-atom laws. They check 108 zero-bias/hinge instances and 324 raw-mixture derivatives, at negative, zero, positive and colliding thresholds. Ninety-eight collisions detect the incorrect closed upper endpoint in the gap measure.

Two further controls identify analytic failure modes. For Bernoulli on $\{-1, 1\}$, the open-event function $y \mapsto \mathbb{P}(X < -y)$ at $y = 1$ has value zero and left limit $1/2$, so it is not upper semicontinuous. For integer $m > 1$, put mass $1/(2m^3)$ at each of $\pm m$, mass $(1 - 1/m)/2$ at each of ± 1, and the remaining mass $1/m - 1/m^3$ at zero. These standardized laws have $\rho_m = 2 - 1/m$, while the weak limit has $\rho = 1$; their third-moment tails at m equal one. This refutes the unjustified UI3 shortcut, not the attainment proof above.

Item	Mathematical status	Execution status
Fixed-index attainment, contact and support	Proved above; support uses its stated moment inputs	Exact finite-law checks pass
Conserved coefficient and sign gates	Exact identities and conditional consequences above	Independent rational algebra passes
Complete Fourier bounds and new local exclusion	Not established by this note	No native run or certificate acceptance